7 The distance from the Earth to the Sun is 1.496×10^{11} metres.
The speed of light is 3×10^{8} metres per second.

(a) Show that, correct to 3 significant figures, light will take 0.139 hours to travel from the Sun to the Earth.

(3)

1 googol is 1×10^{100}

Danesh says,

"When I multiply 1.496×10^{11} by 6.68×10^{9} I get nearly 1 googol because $1.496 \times 10^{11} \times 6.68 \times 10^{9} = 9.99 \times 10^{99}$"

(b) Is Danesh correct?
Give a reason for your answer.

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(1)

(Total for Question 7 is 4 marks)